

Morphometric Ratios Data Collection Protocol

Miram R. Gleiber^{1,2}, Natasha A. Hardy¹, Cindy Matuch¹, Iris George¹, Alana Krug-Macleod¹, Caitlin J. Morganson¹, Larry B. Crowder³, Stephanie J. Green¹

¹ Department of Biological Sciences, University of Alberta, Edmonton, Alberta, Canada

² Department of Integrative Biology, Oregon State University, Corvallis, Oregon, USA

³ Hopkins Marine Station, Department of Biology, Stanford University, Pacific Grove, California, USA

Purpose: Build a data library of body shape ratios (total length : body height, standard length : body height, total length : total height, standard length : total height, standard length : total length, eye diameter : total length) for juvenile and adult life stages of pelagic species in the California Current, as well as for albacore tuna prey species using images collected online. A data collection spreadsheet is included in the database files (Table 10).

Additional documents:

- *Morphometric Measurement Guide* (Figures 1-10)
- *10_morphometric_ratios_data_collection.csv*
- *10metadata_morphometric_ratios_data_collection.csv*
- *Pelagic_Species_Trait_Database.Rmd*

Image search instructions:

1. Using Google Images, search the species “scientific name” or “common name” and life stage (e.g. “adult”, “juvenile”).
2. Select six images of each life stage per species, if possible, using the following criteria for specimen image selection:
 - a. Images ideally have all dimensions needed for measurements visible on the same image (Figure 1-10). Specimen is depicted showing body length and height (longest and second longest axes), as well as eye and fins (if present). Generally lateral (side) view images are best, however for some organisms dorsal (top) view shows body length and height better (e.g. flatfish, rays, crabs; Figure 6, 9). For soft-bodied species (e.g. cephalopods) collect measurements from images with arms extended (2, 3, 10).
 - b. The organism’s body should be perpendicular to the camera.
 - c. Image depicts the correct species; check the image source information to confirm.
 - d. Photos (professional or amateur) or drawings from literature sources (e.g. textbooks, ID guides). Only use crude drawings if no other options are available. Use the *image_type* column to record if the image is an amateur photo, scientific photo, or sketch.
 - e. Images do not need a scale bar, since we are using relative body measurements.

3. If the image selection is limited (e.g. rare species, juveniles), images might have to be used that don't meet all the above criteria. See the *image measurement troubleshooting* section (below) for details on noting this in the data collection spreadsheet.
4. Name the image file with the species scientific name, life stage, and photo number (e.g. genus_species_adult/juve_#.jpg).
5. Record the image name (*file_name*) and *source* (e.g. Fishbase) in the data collection spreadsheet (Table 10).

Data collection instructions:

1. Download and install ImageJ for your operating system:
<https://imagej.nih.gov/ij/download.html>
2. Open the file in ImageJ and zoom in so the image fills the screen (improves precision of measurements).
3. **Measuring images:** Use the line tool to create a line that represents each measurement. For a curved line, right click the line tool and select 'segmented line'. Use the measurement descriptions and illustrations in the '*Morphometric Measurement Guide*' (Figure 1-10) for detailed instructions on measuring different taxa.
4. Click 'Analyze' -> 'Measure', or type 'M' for a shortcut. The line measurement is in the 'Length' column of the Results pop up window. Lengths measurements can be copied and pasted into the spreadsheet.
5. Record all data in the spreadsheet. Double check measurement values are correct.
6. If there are issues with the specimen in the image that may impact the measurement see the *image measurement troubleshooting* section (below) and include details in the *status_type* and *fin_arm_status* columns. Any notes about the image can also be detailed in the *notes* column.
7. Use the *scale_bar* column to record if the image includes a scale bar, or has a length measurement on the image. We are not using the scale bar in analysis, but noting if present.
8. Close image and go to the next one.
9. After measuring all images for a species and lifestage, decide if each image is appropriate to use in the database and note in the *use* column. Enter the total number of images to be used (*use* = "yes") in the *replicates* column. An example of an image not used could be a photo of a badly decomposed specimen that would distort relative measurements.

Image measurement troubleshooting:

1. Standard length (SL; *s_length*)

- **Image position:** organism at an angle in the image that skews the measurement.
- *s_length* = NA; *status_type* = position; *fin_arm_status* = angle
- **SL no different than TL:** if no difference between SL and TL (e.g. jellies, salps, cladocerans); enter the same value for both.

2. Total length (TL; *t_length*)

- **Not visible:** part of the body, arms or fins are not visible in the image (e.g. shell only for gastropods, body tucked in shell, cephalopod with arms not extended.)
 - *t_length* = NA; *status_type* = arms, body; *fin_arm_status* = not visible, folded
- **Image position:** organism at an angle in the image that skews the measurement.
 - *t_length* = NA; *status_type* = position; *fin_arm_status* = angle
- **TL no different than SL:** if no difference between SL and TL (e.g. jellies, salps, cladocerans); enter the same value for both.

3. Body height (BH; *b_height*)

- **Image position:** organism at an angle in the image that skews the measurement (e.g. dorsal view image for fish, lateral view image for crabs)
 - *b_height* = NA; *status_type* = position; *fin_arm_status* = angle or top

4. Total height (TH; *t_height*)

- **Not visible:** organisms (e.g. fish, cephalopods) with fins folded or not visible*
 - *t_height* = NA; *status_type* = fins; *fin_arm_status* = not visible, folded
- **Image position:** organism at an angle in the image not allowing fins to be seen.
 - *t_height* = NA; *status_type* = position; *fin_arm_status* = angle, side, top
 - Might be able to estimate this as BH + FH + FH if you can see one fin and both are symmetrical, or one fin is only slightly folded; include in *notes*.
- **No fins:** for organisms without fins (octopus, crustaceans, jellies, zooplankton) there is no difference between TH and BH; enter the same value for both.

5. Dorsal fin height (FH; *f_height*)

- **No fins:** organisms without fins (octopus, crustaceans, jellies, zooplankton)
 - *f_height* = 0
- **Not visible:** organisms (e.g. fish, cephalopods) with fins folded or not visible*
 - *f_height* = NA; *status_type* = fins; *fin_arm_status* = not visible, folded
- **Image position:** organism at an angle in the image not allowing fins to be seen
 - *f_height* = NA; *status_type* = position; *fin_arm_status* = angle, side, top

6. Eye diameter (*eye_diameter*)

- **No eye:** organisms without eyes (e.g. some crustaceans, some gastropods, jellies)
 - *eye_diameter* = 0
- **Not visible:** organisms where eye is not visible
 - *eye_diameter* = NA; *status_type* = eye; *fin_arm_status* = not visible

See columns *status_type*, *fin_arm_status*, or *notes* in the individual image morphometric data (Table 5) for examples of recording image measurement issues.